Data Structures and Application – BCS304

Question Bank

Table of Contents

Module - 1 1
Module 1 - Numericals 1
Module – 2 2
Module 2 - Numericals 2
Module – 3 3
Module 3 - Numericals 3
Module – 4 5
Module 4 - Numericals 5
Module – 5 7
Module 5 - Numericals 7

Module - 1

1. What is data structure? Explain the classifications of Data structures with example.
2. What is a structure? Explain the different ways of representing structure with suitable example.
    a. Differentiate structure and array with suitable example.
    b. Differentiate structure and union with suitable example.
3. Define pointers. Explain declaration and initialization of pointers.
4. Explain dynamic memory allocation.
5. C-function to perform polynomial addition.
6. C-function to perform fast Transpose of a sparse matrix.
7. Explain Knuthh Morris Pattern Matching Algorithm.
8. Define a stack. Explain the operations on the stack using suitable C- functions.
9. Write C- function to convert an infix expression to postfix expression.
10. Write C- function to evaluate a postfix expression

Module 1 - Numericals

1. Represent the given sparse matrix in the triplet representation method.
2. Represent the given polynomial in the array.
3. Search pattern “abcdabcy” in the text “abcxabcdabxabcdabcdabcy”. Form the LPS array of KMP
4. Convert the following infix expression to postfix expression(parenthesis method and stack method)
5. Given postfix expression ABC-D*+E$F+ assume A=6,B=3,C=2,D=5,E=1 and F=7.

Module – 2

1. Define a queue. Explain the operations on the queue using suitable C- functions.
2. Define a Circular queue. Explain the operations on the circular queue using suitable C- functions. List the advantages of circular queue.
3. Write a C- function to add two polynomials using linked list.
4. Explain linked stack with suitable C- functions
5. Explain linked Queue with suitable C - functions
6. Write C- functions for the following on a Singly linked list:
    a. Insert a node in the beginning of a list
    b. Insert a node at the end of a list
    c. Insert a node after a node in a list
    d. delete a node in the beginning of a list
    e. delete a node at the end of a list
    f. delete a node after a node in a list
    g. Search for a node
    h. Print the list
    i. Count the number of nodes in the list
    j. Concatenate two list

Module 2 - Numericals

1. **Represent the following matrix in the linked representation form**

$$\begin{bmatrix} 0 & 5 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Programs will be asked in Linked list.

Module – 3

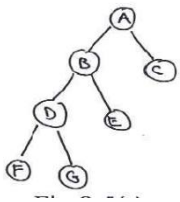
1. What are the advantage of DLL over SLL.
2. **C- function to perform insertion and deletion of an element in a DLL(very important)**
3. C- function to perform polynomial addition using linked list.
4. What is a tree? Explain the following trees:
    a. Binary Tree
    b. Siblings
    c. Leaf node
    d. Level of the Binary tree
    e. Strictly Binary Tree
    f. Complete Binary Tree
    g. Skewed Binary Tree
    h. Degree of the tree
5. Explain the construction of a threaded binary tree.

Module 3 - Numericals

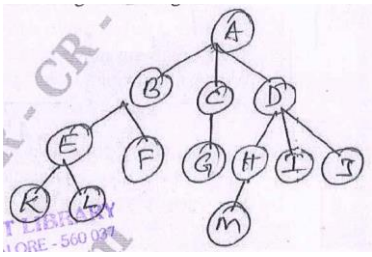
1. For the given sparse matrix, write the diagrammatic linked list representation

$$A = \begin{bmatrix} 0 & 10 & 0 & 0 \\ 3 & 0 & 0 & 5 \\ 8 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 8 & 0 \end{bmatrix}.$$

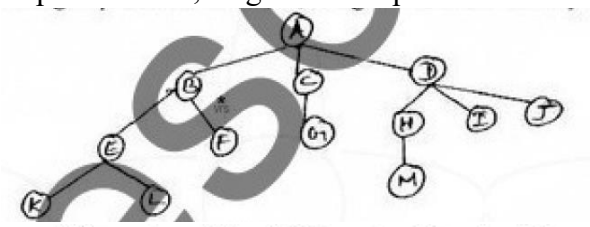
2. Construct Expression tree for the following expression:
    a. 3+4*(7-6)/4+3
    b. a+b*c-d/e*f
    c. AB/C*D*E+
    d.
3. For the following tree find the following:
    a. Siblings
    b. Leaf node
    c. Ancestors
    d. Depth of a tree
    e. Level of a tree



4. Construct a threaded binary tree for 10,20,30,40,50.
5. For the tree give the degree, Nonterminal and terminal nodes, siblings, Ancestors, level, Height or depth.



6. Represent the tree using Linked Representation. Left- child Right Sibling Representation, Degree-two representation.



c. Write a note on Threaded Binary tree for a given Binary tree in Fig.Q7(c), Insert 'r' as a right child of 'S' in a Threaded Binary tree and write the function to insert **(10 Marks)**

Fig.Q7(c)

7.

Module – 4

1. Define BST. Write C – function to insert element into a BST
2. Write the recursive and iterative search for a BST.
3. Write algorithm for deleting an element in a BST.
4. C- function to traverse a Tree
    a. Preorder
    b. Postorder
    c. Inorder
5. What is a selection tree? Explain the types with suitable examples.
6. Explain the union and find operation on a set and how it is represented using a tree.
7. Explain the collapsing rule.
8. Define the following with respect to a graph:
    a. Connected Graph
    b. Directed Graph
    c. Multigraph
    d. Complete Graph
    e. Subgraph
9. What are the different graph traversal? Explain with suitable C- functions.
10. Explain Spanning tree of a graph.
11. What is articulation point in a graph. Explain with suitable examples.

Module 4 - Numericals

1. For the given data, draw a BST and show the array and linked representation of the same.
    a. 100,85,45,55,110,20,70,65
    b. 1,3,8,5,7,9,10,12,15,14,13,11,6
    c. 4,12,10,18,24,22,15,31,44,35,66,90,70,50,25

2. Traverse the tree using inorder, preorder and postorder Traversal

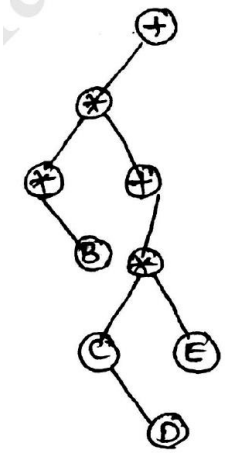
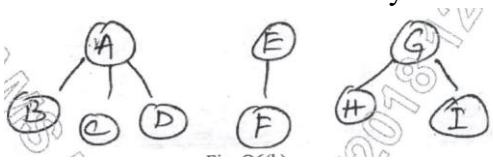


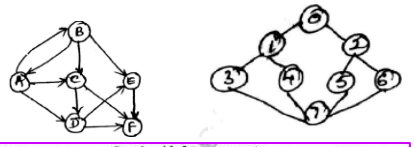
Fig.Q8(b)

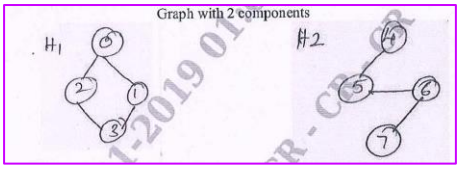
3. Construct a BST by using the following inorder and preorder traversals:
    a. Inorder : BCAEDGHFI and Preorder: ABCDEFGHI
    b. Inorder: DJGBHEAFKIC and Postorder: JGDHEBKIFCA

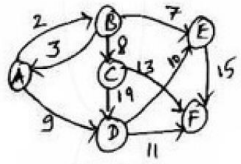
4. Convert the forest into a binary tree:

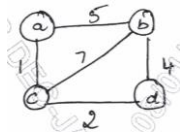


5. Give adjacency matrix and adjacency list for the following graph:

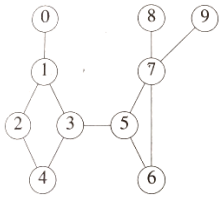








6. Given a graph G. Find the articulation point for the graph



(a) Connected graph

Module – 5

Hashing, Priority Queue, Leftist tree, Optimal Binary Search tree

1. Write Short notes on Hashing Function
2. Explain in detail static and dynamic hashing.
3. What is Collision? Give the different methods to handle collision?
4. Explain Linear Probing with suitable example and C- function.
5. Explain Dynamic Hashing with suitable example.
6. What is a Priority Queue? Explain the different types of priority queues.
7. What is a heap. Explain the different types of heap with suitable example.
8. Explain single ended and double ended priority queue.
9. Explain leftist tree. Explain the operations performed on a leftist tree with suitable example.
10. **Explain the meld operation with suitable example**
11. **Explain Optimal Binary Search Tree with suitable examples.**

Module 5 - Numericals

1. Construct the Leftist tree for the following data: 50,75,25,55,40,65
2. Construct an OBST given keys (a1,a2,a3) = (5,10,15) with equal probabilities pi=qi=1/7.
3. Construct an OBST given keys (a1,a2,a3) = (5,10,15) with probabilities (p1,p2,p3)=(0.5,0.1,0.05) and (q0,q1,q2,q3)=(0.15,0.1,0.05,0.05)
4. Construct a OBST given keys (a1,a2,a3,a4)=(10,15,20,25) with (p1,p2,p3,p4)=(3,3,1,1) and(q0,q1,q2,q3,q4)=(2,3,1,1,1).
5. Initially following keys 10,16,11,1,3,4,23 and 15 are inserted into an empty hash table of length of 10. Using open addressing with hash function h(k)=k mod 10 and linear probing. What is the resultant hash table?